



INDIA.RUNS

Build what next India runs on

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Problem Statement: Intelligent Candidate Discovery - rank candidates the way a strong recruiter would, beyond keyword matching.

Solution Overview

- RecruiterLens is a CPU-only candidate ranking system for Redrob's Senior AI Engineer role.
- It builds an evidence graph from career history, skills, behavioral signals, logistics, and risk flags.
- Career evidence outranks keyword overlap: shipped search, ranking, retrieval, and recommendation work matters most.
- Trust-weighted skills use proficiency, duration, endorsements, and assessment scores.
- Risk gates suppress keyword stuffers, services-only profiles, CV/speech-only profiles, and honeypot-style impossibilities.

JD Understanding & Candidate Evaluation

- Target profile: 5-9 years, senior IC judgment, strong Python, product-company ML depth, and shipping bias.
- Must-have evidence: production retrieval/search/ranking/recommendation systems, evaluation frameworks, and vector or hybrid search exposure.
- Important logistics: Pune/Noida preferred, Indian tier-1 cities welcome, hybrid/flexible availability, and shorter notice periods.
- Negative evidence: pure research, recent framework-only LLM demos, non-technical current roles, services-only careers, and unsupported skill stuffing.

Ranking Methodology

- Stream candidates from JSONL and compute deterministic feature panels per profile.
- Score career evidence, current role fit, trusted skill evidence, experience band, company context, logistics, and behavioral availability.
- Apply hard caps when evidence is missing: skills alone cannot make a candidate top-tier.
- Sort by final score with deterministic tie-breaks, then generate one fact-grounded reason per selected candidate.

Explainability & Data Validation

- Every row includes a concise reason built only from observed candidate fields: title, years, evidence, response rate, notice, and location.
- The diagnostics file exposes component scores and risk flags for auditability.
- No LLM generates candidate reasoning during ranking, so there are no hallucinated skills or unsupported claims.
- Official validation passes: exactly 100 rows, ranks 1-100, unique IDs, and non-increasing scores.

End-to-End Workflow

- Read JD and candidate schema, then encode the role as a recruiter-lens scoring rubric.
- Stream 100,000 JSONL profiles through feature extractors and risk gates.
- Rank top 100 candidates and emit CSV with candidate_id, rank, score, and reasoning.
- Validate with the bundled challenge script; export a matching XLSX for the portal.

System Architecture

JD Lens -> structured requirements: must-haves, nice-to-haves, negative evidence, logistics

Candidate Parser -> career roles, descriptions, skills, education, and Redrob behavioral signals

Evidence Scorers -> career fit, role fit, trusted skills, product context, experience, logistics, behavior

Risk Gates -> keyword stuffing, non-technical current role, services-only path, impossible timelines

Ranker + Reasoner -> deterministic top-100 CSV/XLSX with audit-ready candidate explanations

Results & Performance

- Full 100,000-candidate run completes in about 90 seconds on local CPU in this workspace.
- The official CSV validator passes without format, rank, duplicate, or score-order issues.
- Top results are dominated by candidates with shipped ranking/search/retrieval/recommendation systems, not generic AI keyword lists.
- The ranker uses no network, hosted APIs, GPU, or model downloads during inference.

Technologies Used

- Python standard library for the reproducible ranker: json, csv, datetime, regex, dataclasses, and unittest.
Deterministic evidence dictionaries and risk gates instead of per-candidate LLM calls.
Artifact-tool used only for submission artifacts: XLSX formatting and PPTX generation.
No heavy dependencies keeps Stage 3 reproduction simple under the 5-minute CPU/no-network constraint.

Submission Assets

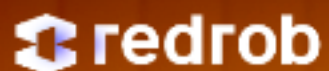
GitHub repository: <https://github.com/aarav06gupta/redrob-recruiter-lens>

Reproduce command: `python rank.py --candidates ./candidates.jsonl --out ./submission.csv`

Ranked outputs: `outputs/redrob_submission.csv` and `outputs/redrob_submission.xlsx`.

Audit files: `outputs/redrob_diagnostics.csv`, `README.md`, `tests`, and `submission_metadata.yaml`.

Sandbox link: TODO - deploy the same command in Hugging Face Spaces, Streamlit Cloud, Replit, Colab, Docker, or Binder.



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THANK YOU